
ROCKWOOL SANDWICH PANEL

TECHNICAL INSTALLATION GUIDE

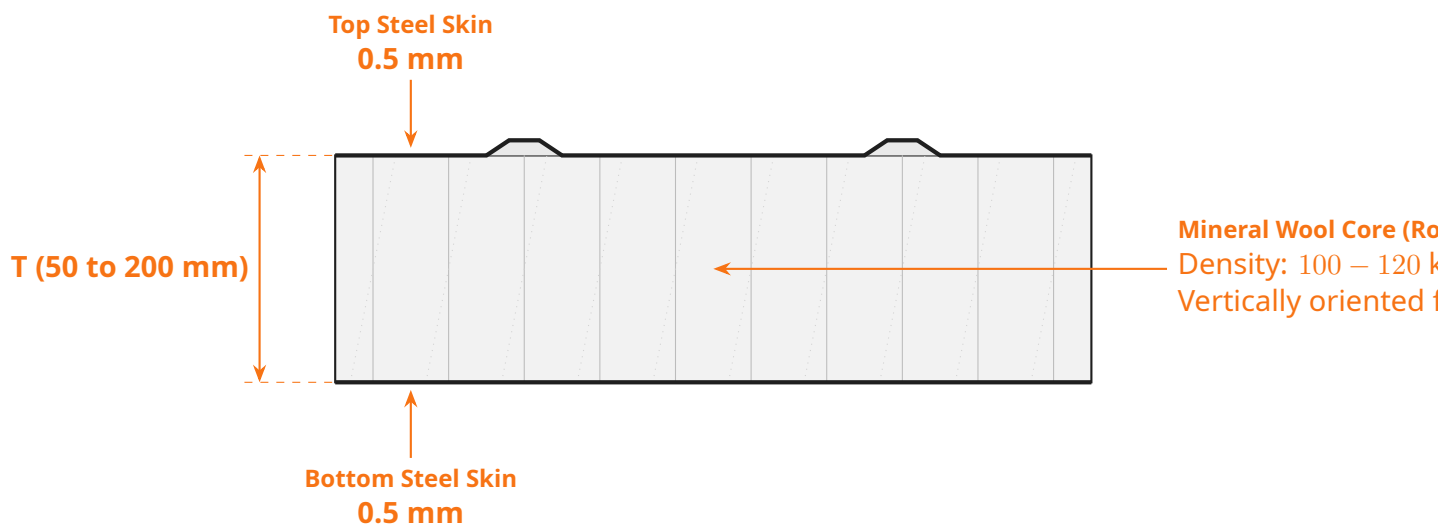
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TECHNICAL SUPPORT AND INDUSTRIAL DESIGN SPECIFICATIONS.
Always ensure you have the latest version before on-site installation.

PANEL CROSS SECTION

System Description

The Rockwool sandwich panel consists of two symmetrically profiled steel skins with a high-density mineral wool fiber core oriented perpendicular to the faces. This design provides exceptional mechanical stiffness combined with outstanding thermal and acoustic insulation properties, as well as high fire resistance.



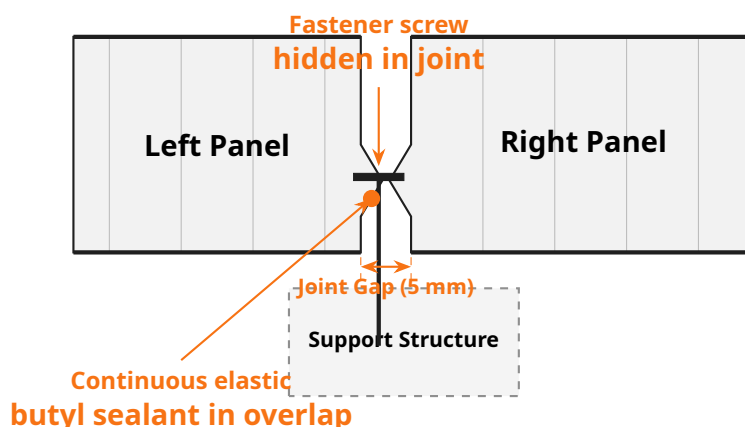
Standard Physical Specifications

- **Effective Coverage Width:** 1000 mm (manufacturing standard).
- **Panel Thickness (T):** Available in 50, 80, 100, 120, 150, and 200 mm.
- **Surface Finish:** Galvanized prepainted steel, smooth or lightly ribbed finish.
- **Fire Rating:** Non-combustible core with fire reaction classification $A1$ or $A2 - s1, d0$.

WALL JOINT DETAIL

Male-Female Tongue and Groove Joint

The longitudinal joint for wall panels utilizes a geometric male-female interlocking design to ensure a tight seal. To guarantee airtightness and moisture protection, a continuous bead of non-hardening sealant (butyl or equivalent) must be applied prior to interlocking the wall joint during alignment.



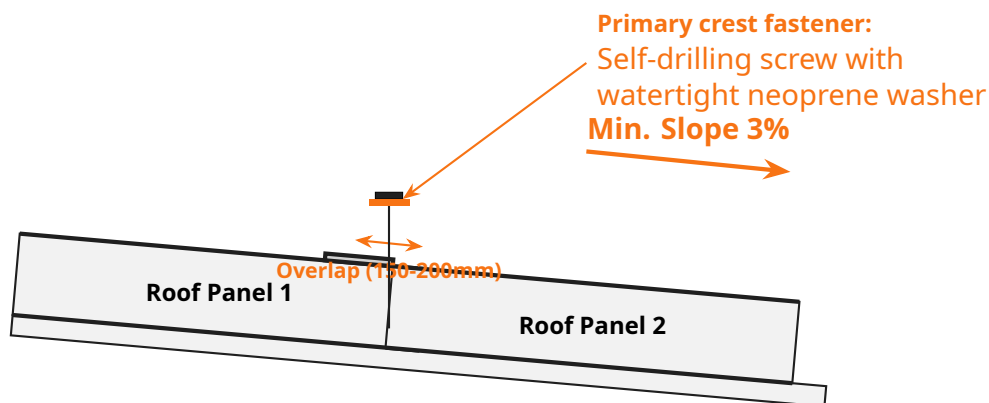
Installation Instructions

1. **Alignment:** Always verify the verticality of the structure before installing the first panel.
2. **Gasket Application:** Place self-adhesive elastomer tape on the rear steel support member to prevent direct thermal bridging and expansion/contraction noise.
3. **Interlocking Pressure:** Push the second panel firmly against the first, ensuring the tongue engages fully inside the groove cavity.

ROOF PANEL DETAIL

Assembly Layout and Slopes

Installing sandwich panels for roof applications requires strict verification of the minimum project slope (3% minimum for continuous runs without intermediate end laps; 5% to 7% if longitudinal panel end lapping is required). The rainwater on the roof must always flow in the direction opposite to the longitudinal assembly sequence of the panels.



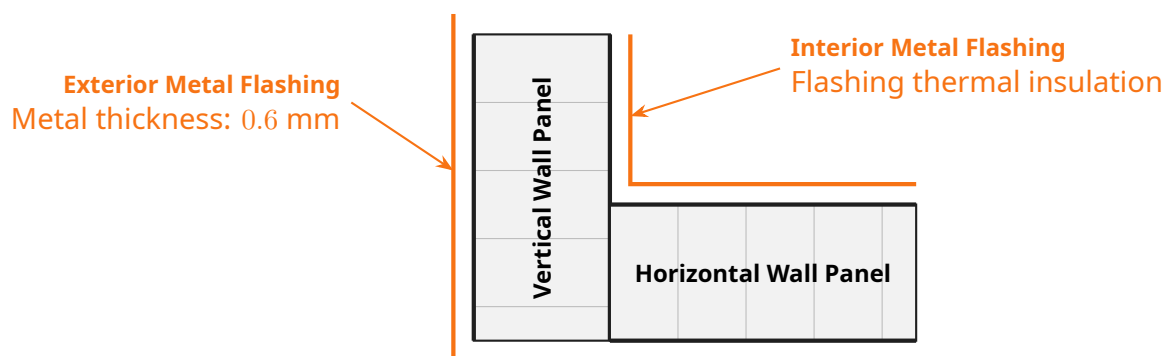
Key Installation Specifications

- **Fastening Line:** Screws must be fastened on top of the panel's crown/profile rib to prevent rainwater pooling and subsequent leakage.
- **Lay Direction:** Installation should progress in the direction opposite to the prevailing winds to mitigate water infiltration under dynamic wind pressure.

CORNER & BASE DETAILS

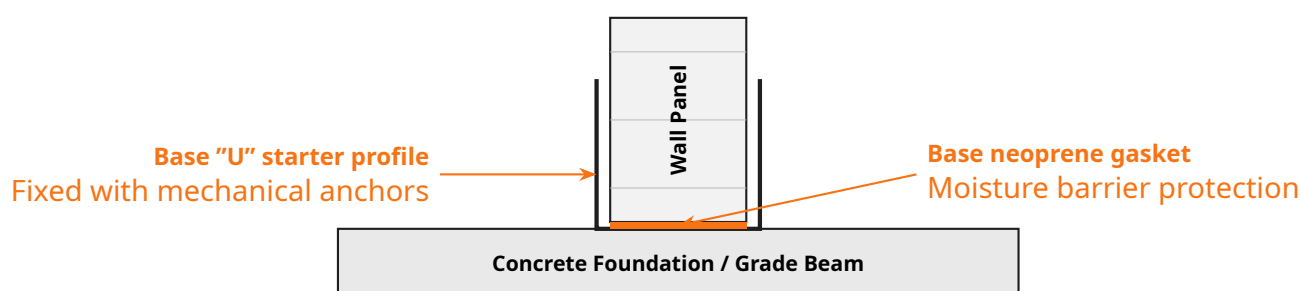
Detail A — Exterior Corner Joint (Horizontal Layout)

The corner intersection is resolved using an angled miter cut on the panels or by installing structural prepainted interior and exterior corner flashings, mechanically fixed to the panel skins. The remaining void must be filled with compressed rockwool or injected polyurethane foam to prevent thermal losses.



Detail B — Base Channel / Starter Profile

To support the first row of wall panels at foundation level, a galvanized steel "U" starter profile must be anchored to the concrete beam at 400 mm intervals.



FASTENER SCHEDULE

Screw Specifications and Spacing Intervals

The correct selection of fastener type, length, and diameter is a determining factor in the structural integrity of the thermo-acoustic building envelope. Fasteners must withstand both suction forces caused by local wind gusts and thermal bowing of the metal faces due to seasonal temperature differentials.

Panel Thickness	Fastener Type	Min. Length	Wall Spacing	Roof Spacing
50 mm	Self-Drilling 6.3 mm	85 mm	1.50 m	1.20 m
80 mm	Self-Drilling 6.3 mm	115 mm	1.40 m	1.10 m
100 mm	Self-Drilling 6.3 mm	135 mm	1.30 m	1.00 m
120 mm	Self-Drilling 6.3 mm	155 mm	1.20 m	0.90 m
150 mm	Self-Drilling 6.3 mm	185 mm	1.10 m	0.80 m
200 mm	Self-Drilling 6.3 mm	235 mm	1.00 m	0.70 m

Table 1: Fastener sizing and recommended span spacing based on generic calculation.

Final Technical Considerations

- **Fastener Material Quality:** The use of martensitic stainless steel screws type A2 or A4 with zinc-nickel protective coating is prescribed for exterior and corrosive environments.
- **Clamping Force:** The screwdriver torque clutch must be set to compress the EPDM sealing washer to approximately 30% of its nominal resting thickness. Over-tightening causes local deformation of the outer sheet metal preventing regular runoff, while under-tightening results in seal failure.

END OF TECHNICAL SPECIFICATION — MODEL RW-SP-WD-2024